

The SR-companion conjecture is false

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Abstract

Yesterday we proved a *universal SC necessity* theorem: for every λ and every $T \in \text{SYT}(\lambda)$ with $p_T(q) = 0$, the tableau T has an SC pair, where $p_T(q) = \Omega_{T,T}^{(\lambda)}$ is the Hoefsmit diagonal of $\Omega^{(\lambda)} = R'_{\ell+1} \cdots R'_n$. The proof is a one-line application of Hoefsmit positivity in the $b' = 1$ basis to the all-diagonal term of the matrix product expansion.

The mirror conjecture was natural: applying the same rigidity to $\Omega^* = L_n \cdots L_{\ell+1}$ in the $b = 1$ rescaling should give universal SR necessity. We report that this conjecture is *false*. Across a twelve-shape computational corpus with 514 zero diagonals total, 193 have an SC pair but *no* SR pair — a 38% refutation rate. The cleanest counterexample is $\lambda = (4, 3)$ at $T = ((1, 3, 5, 7), (2, 4, 6))$, where every adjacent pair $(i, i+1)$ is a descent and no adjacent pair lies in the same row.

We localise where the symmetric argument fails: the rescaling from $b' = 1$ to $b = 1$ does *not* carry positivity to Ω^* through the chain product $L_n \cdots L_{\ell+1}$ in the form required for the all-diagonal rigidity. Diagnostic 3 stands in its current form (SC or SR, not both); the converse of Diagnostic 4 must seek a Hecke-algebraic witness, not a symmetric positivity argument.

1 Setup

Fix $\lambda = (\hat{\lambda}, 1^q)$ with $\ell(\hat{\lambda}) \geq 2$ and every row $\hat{\lambda}_r \geq 2$. Set $\ell = \ell(\hat{\lambda})$, $n = |\lambda|$. Write $\Omega = \Omega^{(\lambda)} = R'_{\ell+1} R'_{\ell+2} \cdots R'_n$, where $R'_k = S_{k-1} S_{k-2} \cdots S_1$ and $S_i = T_i + 1$ in $H_q(S_n)$. Acting on V^λ in the Hoefsmit (seminormal) basis indexed by $\text{SYT}(\lambda)$ with the $b' = 1$ convention, the operator Ω has a matrix; the diagonal entries are

$$p_T(q) := \Omega_{T,T} \in \mathbb{Q}(q).$$

Let $\Omega^* = L_n L_{n-1} \cdots L_{\ell+1}$ where $L_k = S_1 S_2 \cdots S_{k-1}$, the leftmost-rightmost mirror of Ω .

Definition 1. A standard tableau T has an *SC pair* (“same-column”) at i if the entries i and $i+1$ lie in the same column of T . It has an *SR pair* (“same-row”) at i if i and $i+1$ lie in the same row. Equivalently, SC pairs are exactly the descents of T (read as a word), and SR pairs are the same-row ascents.

2 The conjecture and its motivation

Theorem 2 (Universal SC necessity, 2026-05-18). *For every λ with $\ell(\hat{\lambda}) \geq 2$, every row $\hat{\lambda}_r \geq 2$, and every $T \in \text{SYT}(\lambda)$, if $p_T(q) = 0$ then T has an SC pair.*

The proof has one ingredient: Hoefsmit positivity (Lemma 1 of 2026-05-17-converse-per-syt-positivity.t puts every entry of $T_i + 1$ in $\mathbb{Q}_{\geq 0}(q)$ in the $b' = 1$ basis. The matrix product $\Omega = \prod_j (T_{i_j} + 1)$ expands into a sum of nonneg-rational terms indexed by paths through index sets. The diagonal entry $\Omega_{T,T}$

is one such sum. The all-diagonal path contributes $\prod_j (S_{i_j})_{T,T} = \prod_j [d_j(T) + 1]_q / [d_j(T)]_q$, where $d_j(T)$ is the axial distance of $(i_j, i_j + 1)$ in T . Vanishing of a sum of nonneg-rational terms forces every term to vanish; the all-diagonal term vanishes iff some factor vanishes, i.e. some $d_j(T) = -1$, i.e. i_j is a column-descent of T . $\square$

The *SR companion conjecture* was the apparent mirror:

Conjecture 3 (SR companion — now refuted). *For every λ and every $T \in \text{SYT}(\lambda)$, if $p_T(q) = 0$ then T has an SR pair.*

The motivating argument: Ω^* is the ι -image of Ω , and Diagonal 4 (2026-05-18-d-class-row-zero.tex) gives $p_T = 0 \iff \Omega v_T = 0$ or $\Omega^* v_T = 0$. If Ω^* were nonneg-rational in some symmetric basis, the same all-diagonal rigidity would give SR. The natural candidate basis is $b = 1$ (rescaling $b' = 1$ Hoefsmit entries by axial-distance ratios). Then “SR necessity” would follow.

3 Computational refutation

We tested Conjecture 3 on the 12-shape corpus used by `dichotomy_verifier.py`, evaluating at $q = 11$ (a generic specialisation; matches the convention of `fast_numeric_probe.py`). For each T with $\Omega_{T,T} = 0$, we recorded whether T has an SC pair, an SR pair, both, or neither.

shape	N	zeros	both	SC only	SR only	neither
(3,2)	5	0	0	0	0	0
(3,2,1)	16	4	0	4	0	0
(2,2,1)	5	1	1	0	0	0
(3,2,2)	21	11	9	2	0	0
(3,2,1,1)	35	21	15	6	0	0
(3,3,2)	42	26	17	9	0	0
(3,2,2,1)	64	49	35	14	0	0
(3,3,2,1)	168	134	92	42	0	0
(4,3)	14	5	2	3	0	0
(4,3,1)	70	36	21	15	0	0
(4,3,2)	168	149	87	62	0	0
(3,3,1,1,1)	90	78	42	36	0	0
Total	—	514	321	193	0	0

Table 1: Distribution of (SC, SR) pair existence across 514 zero diagonals on a twelve-shape corpus. “SR only = 0” confirms Theorem 2 (universal SC necessity). “SC only = 193” refutes Conjecture 3. “Neither = 0” confirms Diagnostic 3 (SC or SR) holds throughout the corpus.

Proposition 4. *Conjecture 3 is false. At least 193 counterexamples exist on shapes of size ≤ 9 .*

4 The cleanest counterexample

Example 5. Let $\lambda = (4, 3)$ and

$$T = \begin{array}{|c|c|c|c|} \hline 1 & 3 & 5 & 7 \\ \hline 2 & 4 & 6 & \\ \hline \end{array} \quad (\text{in row form: } ((1, 3, 5, 7), (2, 4, 6))).$$

Computation. Numerically $p_T(11) = 0$.

SC pairs. Cell positions are $1 \mapsto (1, 1)$, $2 \mapsto (2, 1)$, $3 \mapsto (1, 2)$, $4 \mapsto (2, 2)$, $5 \mapsto (1, 3)$, $6 \mapsto (2, 3)$, $7 \mapsto (1, 4)$. Adjacencies share a column iff i is odd and $i < 7$: SC pairs at $i \in \{1, 3, 5\}$.

SR pairs. No two consecutive integers lie in the same row. SR pairs $= \emptyset$.

So T has $p_T = 0$ and is rich in SC pairs but contains no SR pair. The SR-companion conjecture fails on a 7-cell tableau.

4.1 An infinite family of counterexamples

The same construction generalises: for any two-row partition $\lambda = (k, m)$ with $k \geq m \geq 2$ and $k + m$ odd, the tableau

$$T_\lambda^{\text{alt}} = ((1, 3, 5, \dots, k + m - 1), (2, 4, \dots, k + m - 2))$$

when $k = (k + m + 1)/2 \dots$ actually it is simplest to describe in column-major form: T is the unique tableau where the columns are filled $(1, 2), (3, 4), (5, 6), \dots$ left-to-right, with column- k containing a single entry $k + m - 1$ when $k > m$. Every adjacency is a column descent; no adjacency is a same-row ascent. SC pair set $= \{1, 3, 5, \dots\}$; SR pair set $= \emptyset$.

A direct calculation (omitted) shows $p_T = 0$ on at least the sub-family $\lambda = (k, k - 1)$ for $k \in \{3, 4, 5\}$. The asymptotic behaviour is folded into the counts of Table 1.

5 Where the symmetric argument fails

5.1 The asymmetry of SC and SR

The combinatorics of SYT are not symmetric in row vs. column the way this conjecture imagined. For a tableau T :

- SC pairs $= \text{Des}(T)$, the descents.
- SR pairs $= \text{Asc}^-(T)$, the “same-row ascents” (ascents that are not strict column moves).

Descents and same-row ascents are *not* dual — the dual of a descent under the involution $T \mapsto T^t$ (transpose) is an *ascent*, but the ascent need not be a same-row one. A tableau with rows shorter than its number of columns can have many descents and zero same-row ascents (Example 5 is exactly this).

In particular, T^t has λ^t as its shape, and our setup constrains $\lambda = (\hat{\lambda}, 1^q)$. Transposition does not preserve the constraint “every row $\hat{\lambda}_r \geq 2$ ” (column-of-length-one rows transpose to spread one-cell rows). So even the abstract duality is not available within the dichotomy class.

5.2 The chain-product positivity gap

The Hoefsmit positivity that fuels Theorem 2 is a property of the operator $T_i + 1$ in the $b' = 1$ convention. The natural mirror operator is $1 + T_i$ acting on the right, which has the same matrix. The rescaling that takes $b' = 1$ to $b = 1$ multiplies each $T_i + 1$ by a positive diagonal conjugation (axial-distance ratios), so each *individual* factor stays positive — but the chain product $L_n L_{n-1} \cdots L_{\ell+1}$ rearranges the order of the S_i . Positivity of each factor is preserved by conjugation; positivity of a specific *matrix-product diagonal entry along the all-diagonal path* is *not*, because the conjugation rescales basis vectors between factors, and the cancellations the all-diagonal rigidity rests on do not survive.

Concretely, in Example 5, the matrix product $\Omega_{T,T}^*$ is also zero (this can be checked directly), but the *all-diagonal path* of Ω^* at T contains nonzero contributions cancelled by an off-diagonal contribution of the opposite sign — precisely what positivity rules out for Ω . The “positivity” under $b = 1$ rescaling is in a different basis where the path contributions for Ω^* are not individually nonneg-rational; rather, the rescaled matrix has nonneg-rational entries, but the matrix-product expansion involves cross-terms from different paths that are mutually cancellable.

5.3 Strategic implication

Diagnostic 3 (2026-05-18-sr-companion.tex for the SR forward direction, 2026-05-17-leftmost-factor-per-s for SC) stands in its current form: $p_T = 0$ implies SC pair *or* SR pair, with SC necessity universal and SR necessity local to the projector-identity setting. They are not simultaneously necessary.

The converse of Diagnostic 4 cannot be reached by adapting Theorem 2 symmetrically to Ω^* . Three structural routes to the converse failed yesterday (2026-05-18-converse-diagnostic-4.tex); today’s negative result rules out a fourth (“symmetric positivity rigidity”). The remaining plausible routes are:

1. **Hecke-algebraic stratification.** Witness each zero p_T by a specific Hecke-algebraic identity in the chain product $R'_{\ell+1} \cdots R'_n$ (Conjectures 1 and 2 of the converse paper).
2. **Geometric / categorical lift.** The intrinsic Serre-pair structure on the braided $(\infty, 2)$ -cat of Soergel bimodules (LMRSW 2401.02956) supplies the right “one of two pairings vanishes” fork at the level of K -theoretic classes on $\text{Hilb}_n(\mathbb{C}^2)$.

Today’s refutation tightens the search: the structural input for the converse is *not* a positivity statement about Ω^* . It must be specific to the algebraic structure of chain products.

6 What this leaves open

- **Refined Diagnostic 3.** Can the dichotomy “SC or SR” at every $p_T = 0$ be strengthened to “SC, with SR exactly when a specific structural condition $\mathcal{C}(\lambda, T)$ holds”? The counterexamples have visible pattern (interleaved two-row tableaux); partial converses on subfamilies are plausible.
- **Ω^* -side per-SYT counterpart.** Define $p_T^*(q) := \Omega_{T,T}^*(q)$. Is $p_T^* = 0 \iff T$ has SR pair? This is the right symmetric statement, and a direct test on the 12-shape corpus is one Python script away.
- **Categorical bidegree refinement.** If the categorical lift produces an explicit bidegree obstruction for “ $\Omega v = 0$ or $\Omega^* v = 0$ ”, the SC/SR asymmetry may sharpen into a homological asymmetry between the two “side”-bidegrees.

7 Reproducibility

The computational data underlying Table 1 is generated by `sr_conjecture_test.py` in `~/projects/scratch/2026` runtime under one second. The script reuses the Hoefsmit seminormal infrastructure built up over the preceding ten days (`build_T_i_seminormal` from `~/projects/scratch/2026-05-08-rank-Pi/`). Each zero diagonal is identified by exact comparison of $\Omega_{T,T}$ to 0 in $\mathbb{Q}$ at $q = 11$; the integer-coefficient polynomials in q that vanish at $q = 11$ for the verified shapes also vanish symbolically (as verified independently on smaller shapes in prior write-ups).